BUSINESS PROBLEM DOCUMENT & PROJECT CHARTER

Project Title: Superstore Sales Performance & Profitability BI Dashboard

Prepared For: Executive Leadership, Sales Operations, and Regional Directors

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1. EXECUTIVE SUMMARY

The Superstore retail business operates across diverse customer segments, multiple product categories, and nationwide geographic regions. While the business achieved $1.6M in cumulative sales across 22,000 units sold, executive management lacked centralized, granular visibility into profit leakages, channel adoption, seasonal performance fluctuations, and regional sales distribution.

This project delivers an interactive Business Intelligence solution designed to track year-over-year (YoY) revenue and profitability trajectories, assess customer fulfillment modes, and identify growth and margin-expansion opportunities across categories and states.

2. BUSINESS PROBLEM STATEMENT

• Lack of Unified Reporting: Leadership historically relied on disparate tabular reports, making it difficult to analyze correlation between sales volume ($1.6M) and net profitability ($175K, an ~11% profit margin).

• Margin Vulnerability & Seasonality: Significant monthly profit volatility exists YoY (2019 vs. 2020), with dips in early quarters (February–April) and sharp spikes in Q4 (November–December), necessitating better inventory and promotional planning.

• Fulfillment & Logistics Cost Optimization: The majority of shipments rely on Standard Class ($0.33M), but high-speed fulfillment (Same Day, First Class) represents low volume yet disproportionate logistical operational overhead.

• Cash Flow & Payment Collection Friction: Cash on Delivery (COD) represents 43% of all transactions, introducing risks of delivery rejections, delayed cash realization, and operational handling costs compared to Digital Payments (Online 35%, Cards 22%).

3. PROJECT OBJECTIVES

1. Centralize Performance Visibility: Provide executive-level KPIs (Revenue, Volume, Profit, and Shipping turnaround) updated dynamically across 4 geographic regions.

2. Optimize Sales-to-Profit Ratios: Identify top revenue contributors (e.g., Office Supplies at $0.64M vs. Technology at $0.47M) and assess whether low-margin furniture/sub-categories dilute net margins.

3. Analyze YoY Growth Trends: Track monthly performance shifts between 2019 and 2020 to establish seasonal baseline targets for future forecasting.

4. Mitigate COD Exposure: Identify payment adoption patterns to set a strategy for shifting COD share down from 43% to under 30% through targeted digital checkout incentives.

4. KEY PERFORMANCE INDICATORS (KPIs)

- Total Revenue: $1.6M (High-level business top-line health)
- Total Units Sold: 22K (Inventory turnover and scale of operation)
- Net Profit: $175K (Overall operational bottom line, ~10.9% margin)
- Average Shipping Turnaround: ~4 Days (Fulfillment efficiency & customer SLA)
- Customer Segment Concentration: 48% Consumer (Customer diversification risk profile)
- Cash on Delivery (COD) Share: 43% (Working capital cycle and payment processing risk)

5. SCOPE OF WORK

In-Scope:

- Regional & State Mapping: Spatial visualization of sales and profit density across the West, East, Central, and South regions (prominently California, New York, Texas, Washington).
- YoY Comparative Analysis: Monthly sales and profit trends (2019 vs. 2020).
- Categorical Breakdown: Macro-category analysis (Office Supplies, Technology, Furniture) and top sub-categories (Phones: $0.20M, Chairs: $0.18M, Binders: $0.17M).
- Operational Channels: Evaluation of Shipping Modes (Standard, Second, First, Same Day) and Payment Modes (COD, Online, Card).

Out-of-Scope (Future Enhancements):

- Customer-level churn/retention tracking (CLV / RFM modeling).

• Product return and refund rate root-cause analysis.

• Predictive machine-learning forecasting beyond historical trends.

6. KEY FINDINGS & DASHBOARD INSIGHTS

1. Customer Mix: The Consumer segment dominates at 48%, followed by Corporate (33%) and Home Office (19%). Marketing spend must focus on retention of the Consumer base while cross-selling higher-margin corporate contracts.

2. Regional Disparity: The West (33%) and East (29%) account for over 62% of total sales, while the South (16%) is lagging behind, presenting an under-penetrated market.

3. Product Drivers: Office Supplies ($0.64M) drives the highest gross revenue. In sub-categories, Phones ($0.20M) and Chairs ($0.18M) generate the highest individual volumes.

4. Logistics Preference: Standard Class represents the overwhelming preference (~59% of shipping volume), indicating that customers are price-conscious rather than urgency-driven.

7. STRATEGIC RECOMMENDATIONS

• Incentivize Digital Payments: Implement instant discounts (1–2%) or loyalty points for switching from COD (43%) to Online/Cards to reduce delivery return rates and speed up working capital cycles.

• Capitalize on Q4 Seasonality: Allocate additional marketing and stock buffering in August–October to capture the recurring November–December sales and profit surges observed in YoY trends.

• Regional Expansion in South & Central: Conduct regional price sensitivity and competitor audits in the South (16%) to identify barriers to entry.

• Category Profitability Review: Audit the Furniture category ($0.45M) to ensure bulky shipping costs are not eroding net margins compared to higher-margin Technology items.

8. TECHNICAL SPECIFICATIONS

• BI Tool: Microsoft Power BI

• Visuals Utilized: KPI Cards, Donut Charts, Dual-line YoY Charts, Horizontal Bar Charts, Bing Maps visual, Region Slicers.

• Data Source: Superstore Retail Transactional Dataset.